

SANDBOX PROJECT

- Why modeling a reactor from scratch?
 - Simplifications → learning the workflow and tools
 - Educational (neutronics, thermal-hydraulics, software, design choices)
- Concentric nat.U annular fuel element with D2O moderator

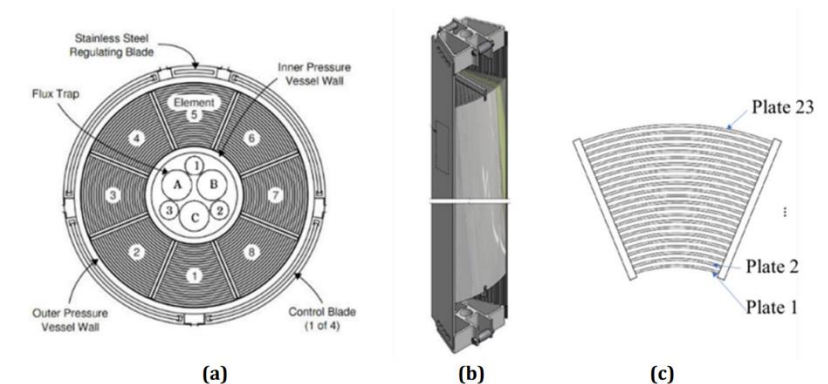
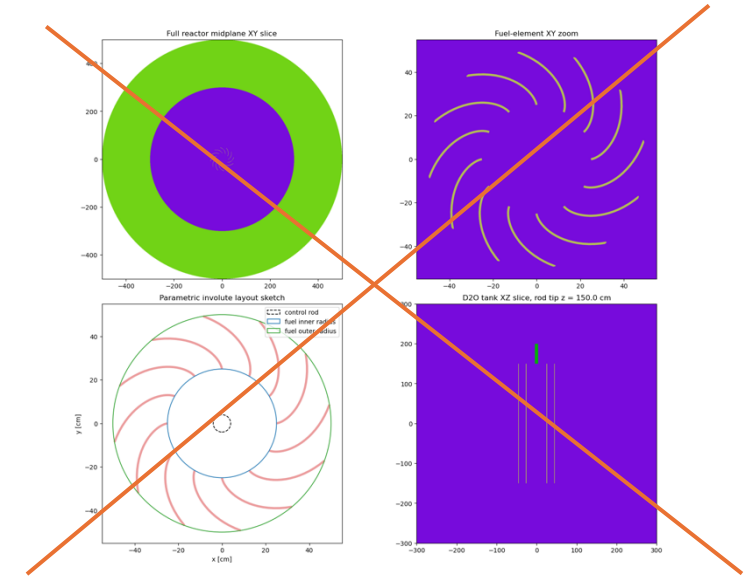
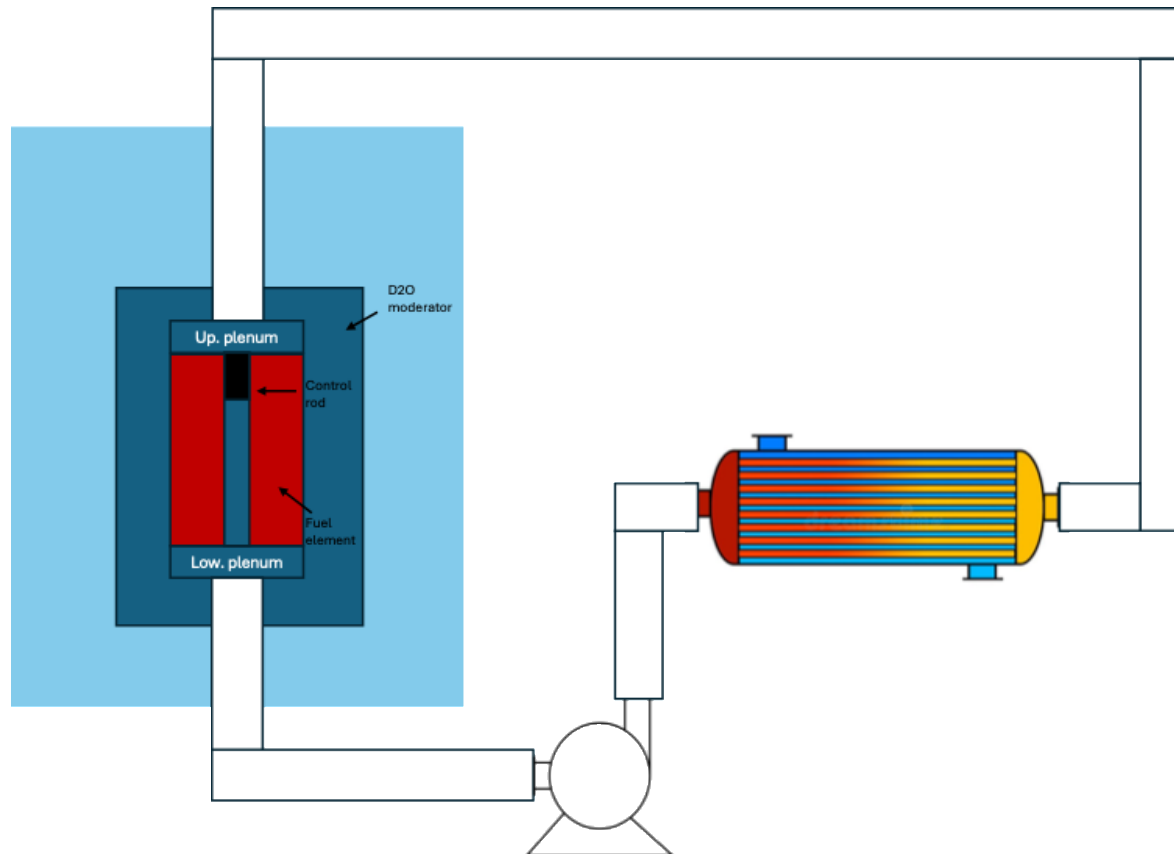
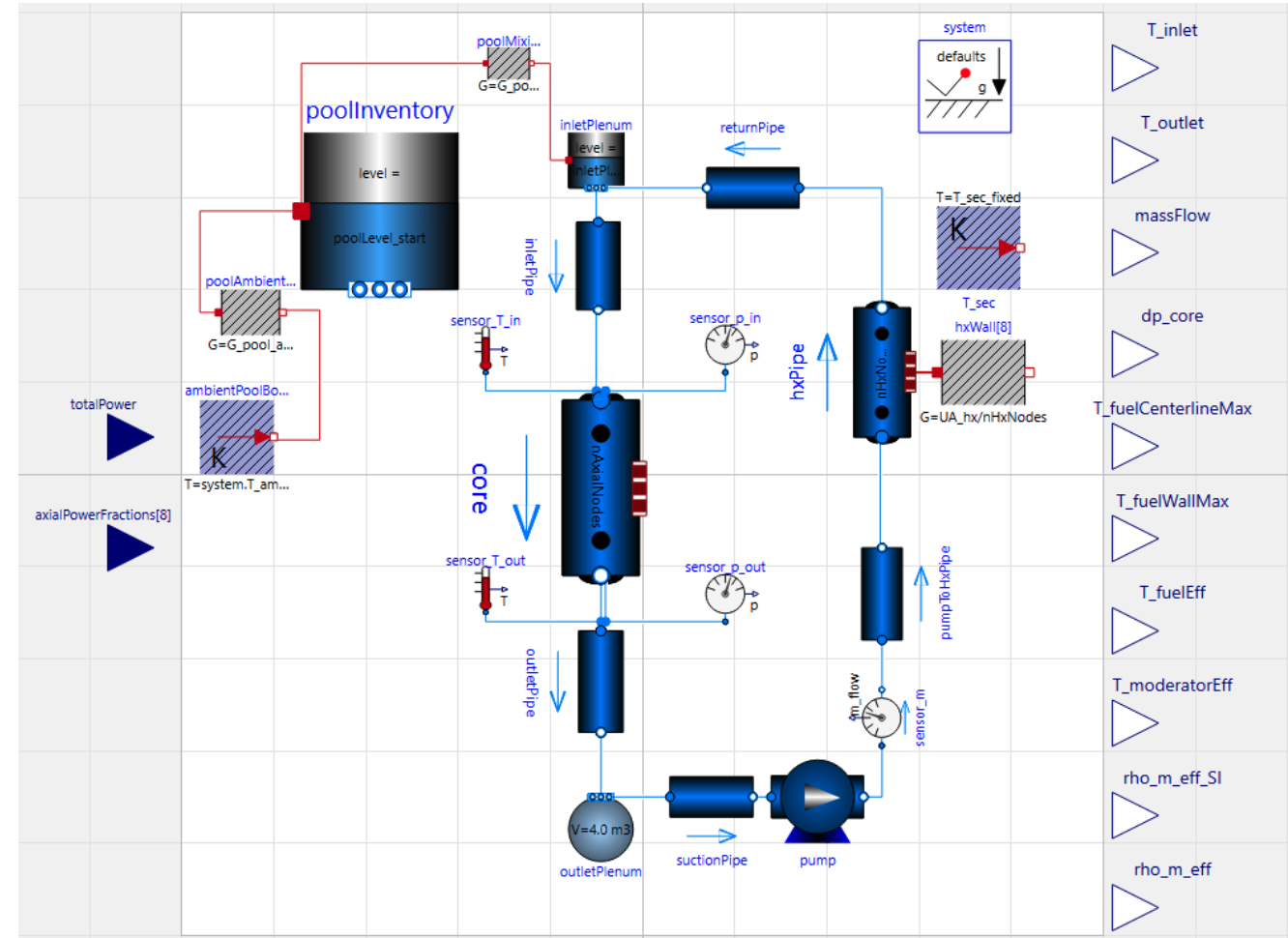


Figure 2.1. MURR core layout (a) and MURR LEU fuel element (b) and cross section (c) [1, 8]

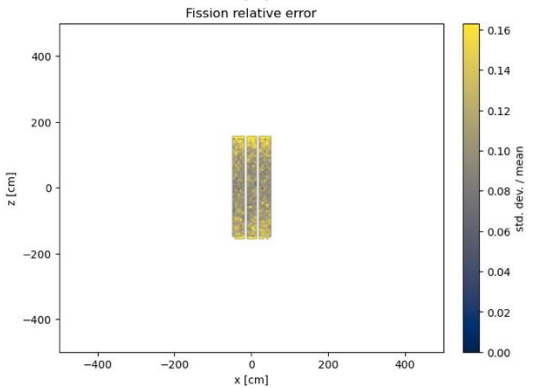
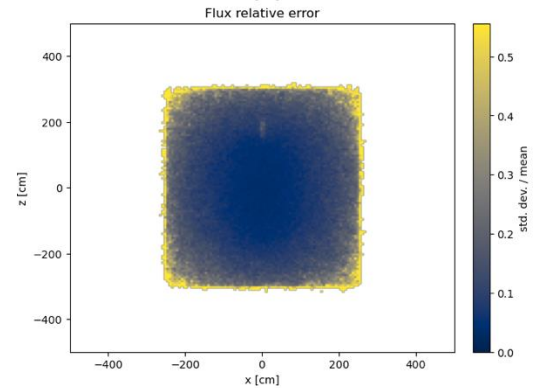
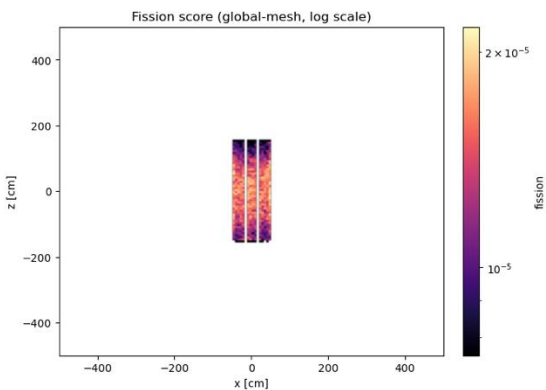
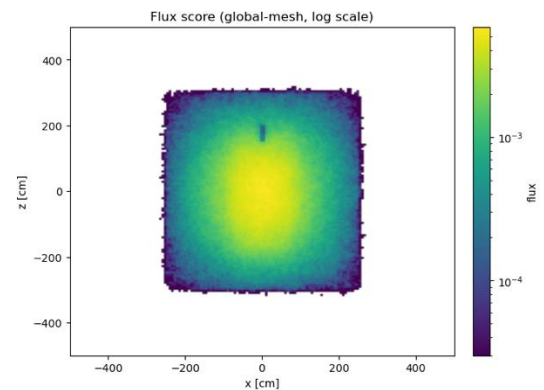
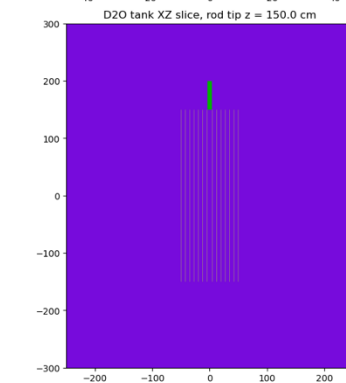
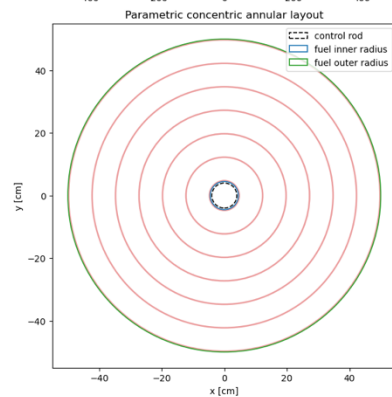
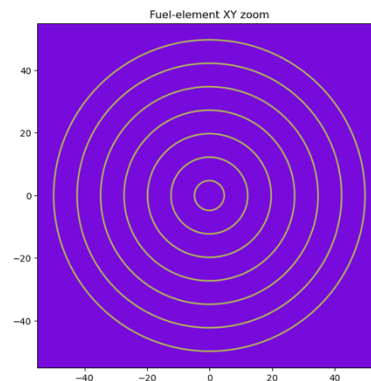
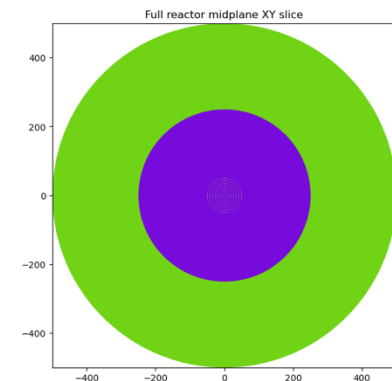
SANDBOX - SYSTEM

- Simple primary loop Modelica model
- Axial nodes for core solving fuel/moderator temp for feedbacks
- At 20MW with 20K ΔT using $\dot{m} = \frac{P}{c_p \Delta T}$
→ 237 kg/s
- Fixed secondary side
- Exported as an FMU and used with fmpy together with the python webapp/solvers



SANDBOX - NEUTRONICS

- Design using OpenMC 
- Fuel element constructor → “physics informed” optimization. $k_{\text{eff}} = 1.00142 \pm 20\text{pcm}$



SANDBOX – ROD WORTH & FEEDBACK

- For a PK model:

$$\rho_{tot} = \rho_{clean} + \Delta\rho_{rod}(x) + \alpha_f \Delta T_f + \alpha_m \Delta T_m + \alpha_{\rho_{D2O}} \Delta \rho_{D2O}$$

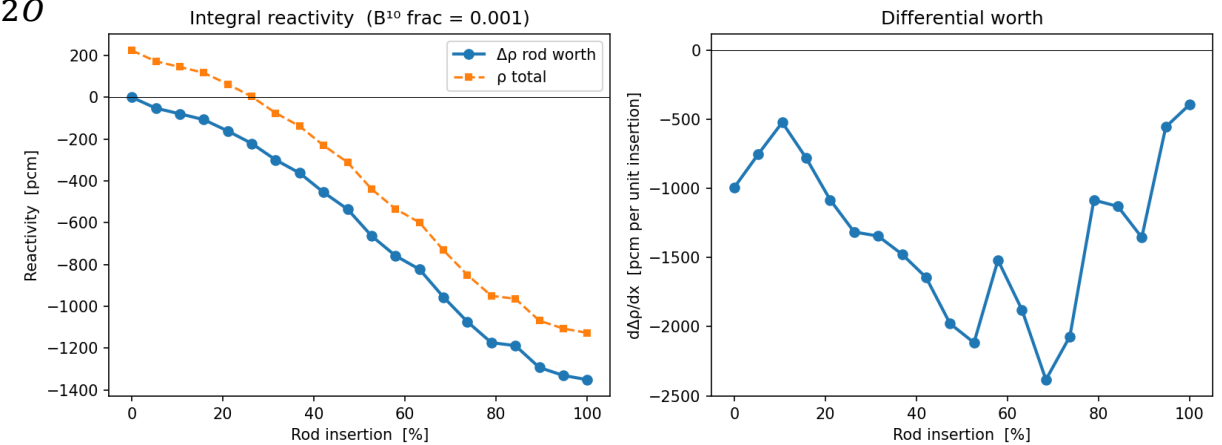
- Running some openmc perturbations:

$$\alpha_f = -2.1383 \text{ pcm/K}$$

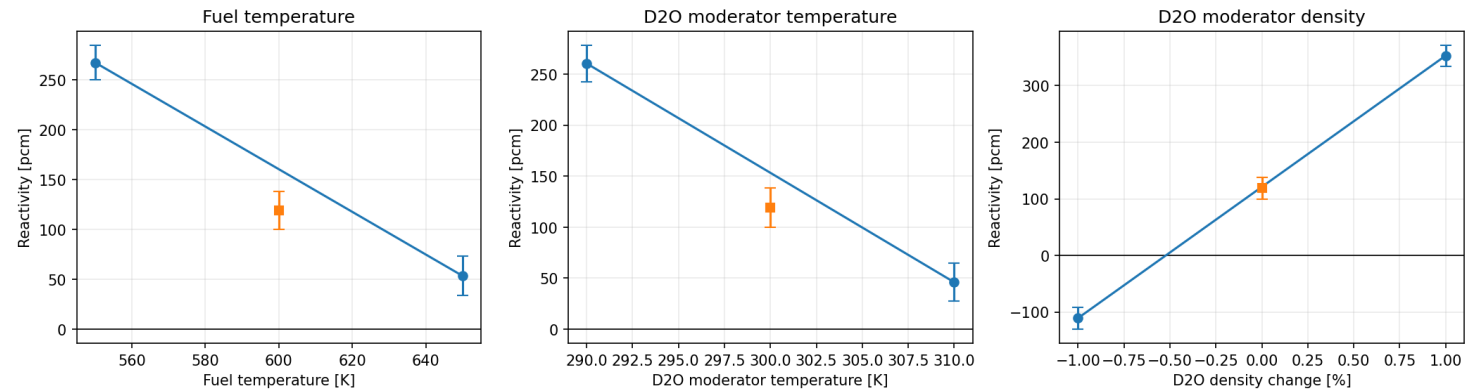
$$\alpha_m = -10.7277 \text{ pcm/K}$$

$$\alpha_{\rho} = 20993 \text{ pcm/(g/cm}^3\text{)}$$

OpenMC rod worth scan (concentric geometry)



Concentric reactor reactivity perturbations



MULTIGROUP COLLAPSE

- Using openmc.mgxs – Multi-Group Cross Section Generation
- Used CASMO group structures from MGXS in openmc
- Divided into regions for each cell (rings, moderator, d2o, h2o tanks).

For a spatial domain V and energy group $[E_g, E_{g-1}]$, the transport-corrected total cross section is calculated as:

$$\begin{aligned}\langle \sigma_t \phi \rangle &= \int_{r \in V} dr \int_{4\pi} d\Omega \int_{E_g}^{E_{g-1}} dE \sigma_t(r, E) \psi(r, E, \Omega) \\ \langle \sigma_{s1} \phi \rangle &= \int_{r \in V} dr \int_{4\pi} d\Omega \int_{E_g}^{E_{g-1}} dE \int_{4\pi} d\Omega' \int_0^\infty dE' \int_{-1}^1 d\mu \mu \sigma_s(r, E' \rightarrow E, \Omega') \cdot \\ \langle \phi \rangle &= \int_{r \in V} dr \int_{4\pi} d\Omega \int_{E_g}^{E_{g-1}} dE \psi(r, E, \Omega) \\ \sigma_{tr} &= \frac{\langle \sigma_t \phi \rangle - \langle \sigma_{s1} \phi \rangle}{\langle \phi \rangle}\end{aligned}$$

<https://docs.openmc.org/en/stable/pythonapi/generated/openmc.mgxs.DiffusionCoefficient.html>

POINT KINETICS

- The MGXS export contains beta and lambda values

$$\frac{dn}{dt} = \frac{\rho - \beta}{\Lambda} n + \sum_{i=1}^6 \lambda_i C_i, \quad (1)$$

$$\frac{dC_i}{dt} = \frac{\beta_i}{\Lambda} n - \lambda_i C_i, \quad i = 1, \dots, 6. \quad (2)$$

$$\frac{n^{k+1} - n^k}{\Delta t} = \frac{\rho - \beta}{\Lambda} n^{k+1} + \sum_{i=1}^6 \lambda_i C_i^{k+1},$$

$$\frac{C_i^{k+1} - C_i^k}{\Delta t} = \frac{\beta_i}{\Lambda} n^{k+1} - \lambda_i C_i^{k+1}.$$

- Rod-worth scan + temperature reactivity coefficient analysis give

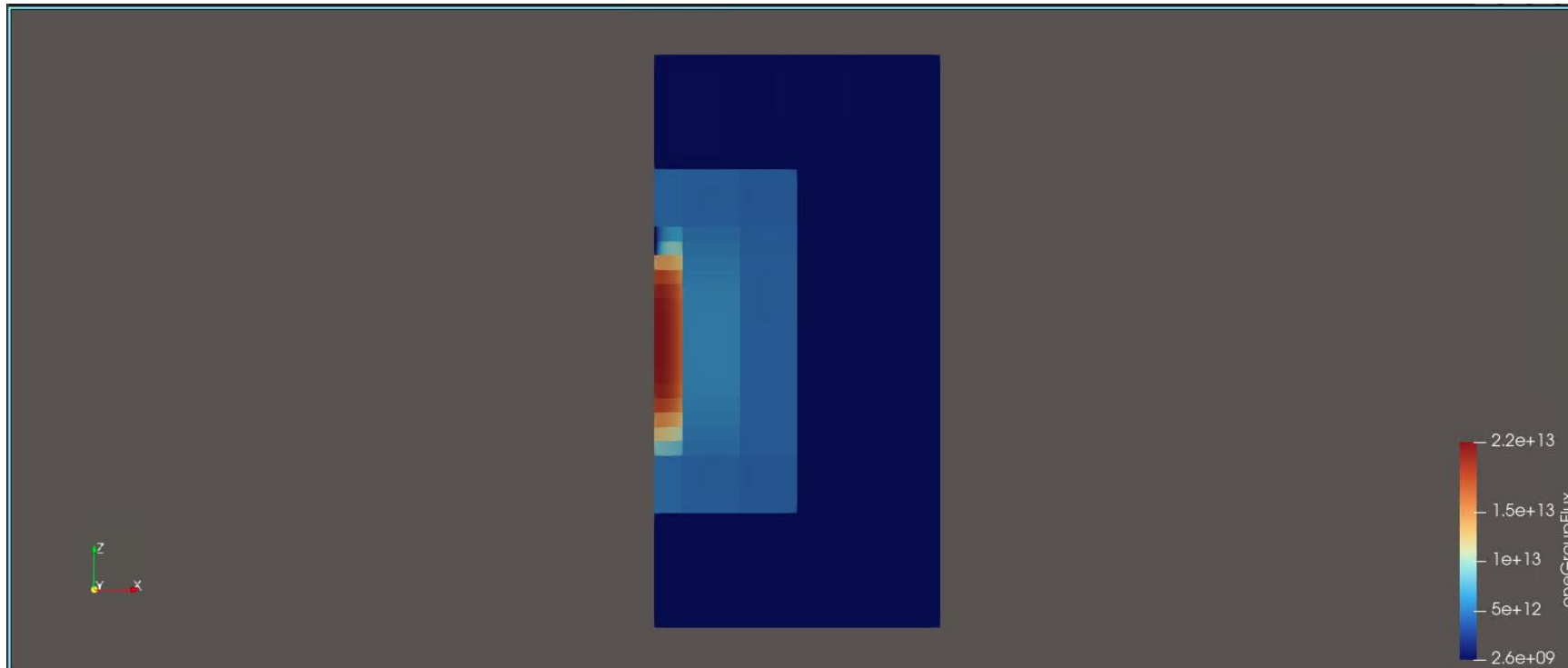
$$\rho_{tot} = \rho_{clean} + \Delta\rho_{rod}(x) + \rho_{feedback}$$

- Python backend solver advances both neutron population and precursors implicitly via:

$$C_i^{k+1} = \frac{C_i^k + \Delta t (\beta_i / \Lambda) n^{k+1}}{1 + \lambda_i \Delta t}, \quad \longrightarrow \quad n^{k+1} = \frac{n^k + \Delta t \sum_i \frac{\lambda_i C_i^k}{1 + \lambda_i \Delta t}}{1 - \Delta t \frac{\rho - \beta}{\Lambda} - \Delta t \sum_i \frac{\lambda_i \Delta t \beta_i / \Lambda}{1 + \lambda_i \Delta t}}.$$

SANDBOX - GENFOAM

- OpenMC MGXS feeds nuclear data but needed some tweaks (P5 without P0 correction).
- GMSH for mesh via python api
- gmshToFOAM app to get neutroMesh
- Keff results: OpenMC keff = 1.00140 +/- 20pcm → GeN-FOAM keff = 0.9765



SANDBOX - DIFFUSION SOLVER

$$-\nabla \cdot (D_g \nabla \phi_g) + \Sigma_{a,g} \phi_g + \sum_{g' \neq g} \Sigma_{s,g \rightarrow g'} \phi_g = \frac{\chi_g}{k_{\text{eff}}} \sum_{g'} \nu \Sigma_{f,g'} \phi_{g'} + \sum_{g' \neq g} \Sigma_{s,g' \rightarrow g} \phi_{g'}.$$

- $D_g = \frac{1}{3\Sigma_{tr,g}}$
- Loss operator $A_g \phi_g = s_g^{\text{fiss}} + \sum_{g' \neq g} \Sigma_{s,g' \rightarrow g} \phi_{g'}$
- Mesh = non-uniform cell-centered finite volume grid in r-Z coordinates.
- Power iteration using CSR matrices solves using `scipy.sparse.linalg.factorized`
- Vacuum boundary condition at extrapolated length per group
- $K_{\text{new}} = k_{\text{old}} * \text{newFission}/\text{oldFission} \mid @\text{volume until converance} < \text{tol} (10^{-5})$

$$k_{\text{new}} = k_{\text{old}} \frac{\sum_{i,j} V_{ij} \sum_g \nu \Sigma_{f,g,ij} \phi_{g,ij}^{\text{new}}}{\sum_{i,j} V_{ij} \sum_g \nu \Sigma_{f,g,ij} \phi_{g,ij}^{\text{old}}} \quad \sum_{i,j} V_{ij} \sum_g \nu \Sigma_{f,g,ij} \phi_{g,ij} = 1$$

POWER CORRECTION

- Rod-induced spatial distortion + baseline correction

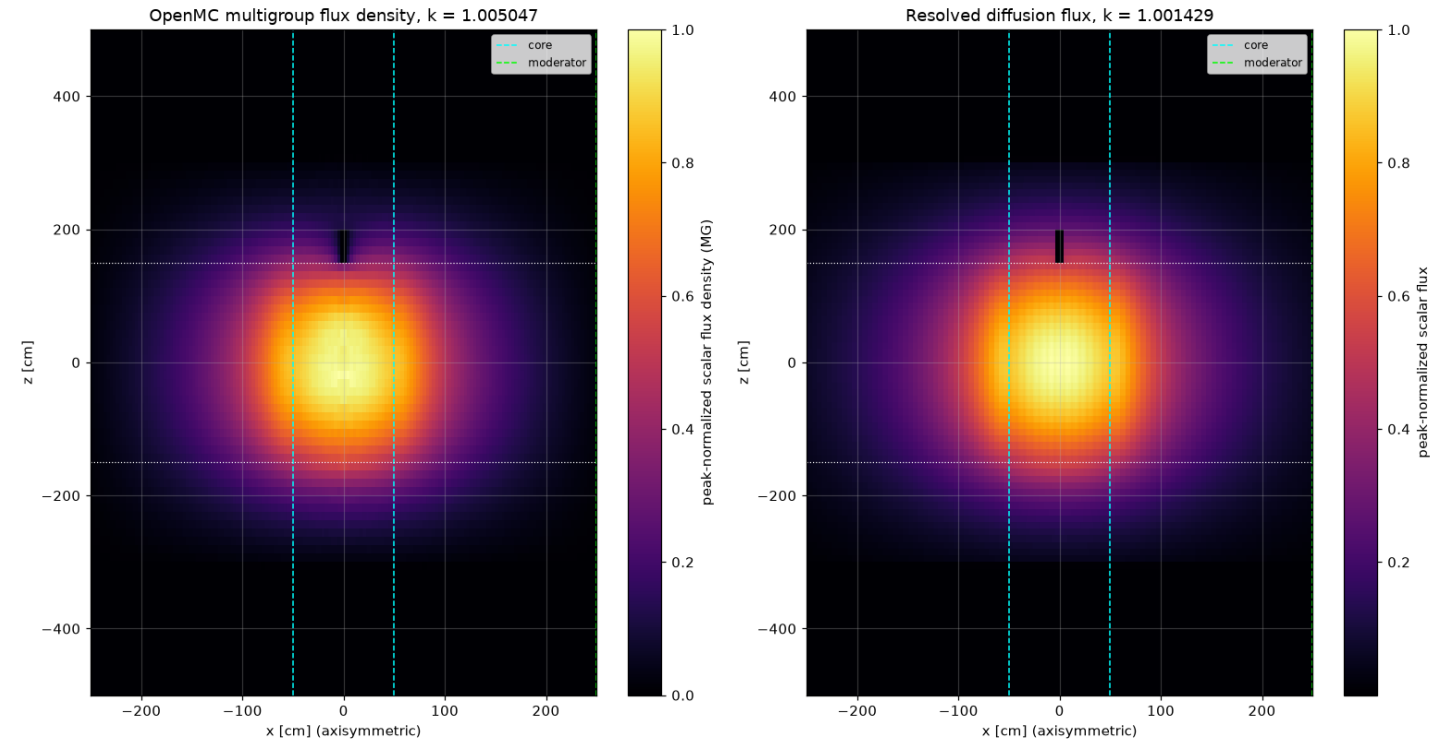
$$C_{clean}(r, z) = \frac{P_{MC}(r, z; x = 0)}{P_{diff}(r, z; x = 0)}$$

$$P_{corr}(r, z; x) = P_{MC}(r, z; x = 0) \frac{P_{diff}(r, z; x)}{P_{diff}(r, z; x = 0)}$$

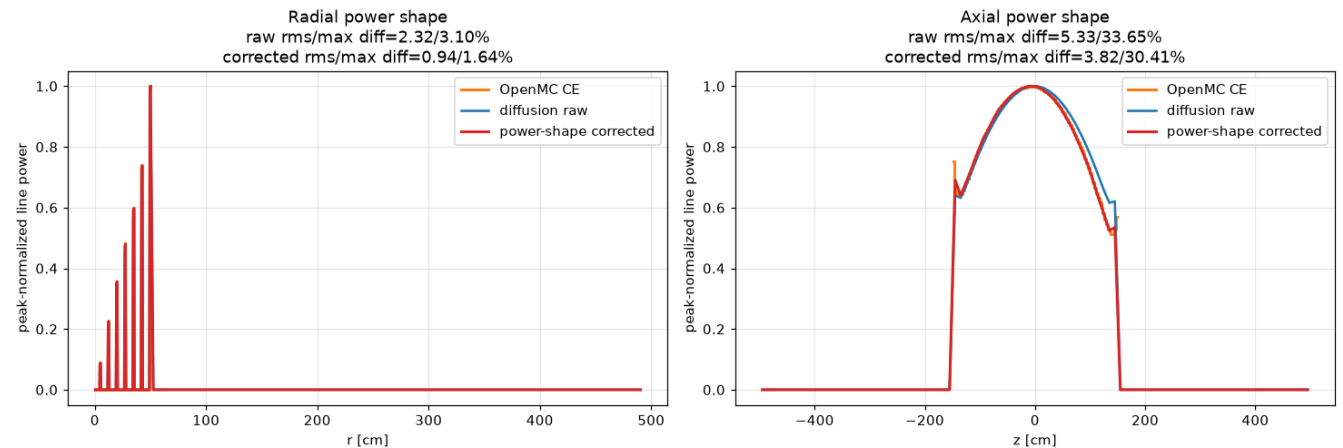
- Then power map uses PK feedback

$$q'''(r, z, t) = P_{PK}(t) \hat{P}(r, z; x)$$

- Power map feeds axial nodes of core model in Modelica TH system

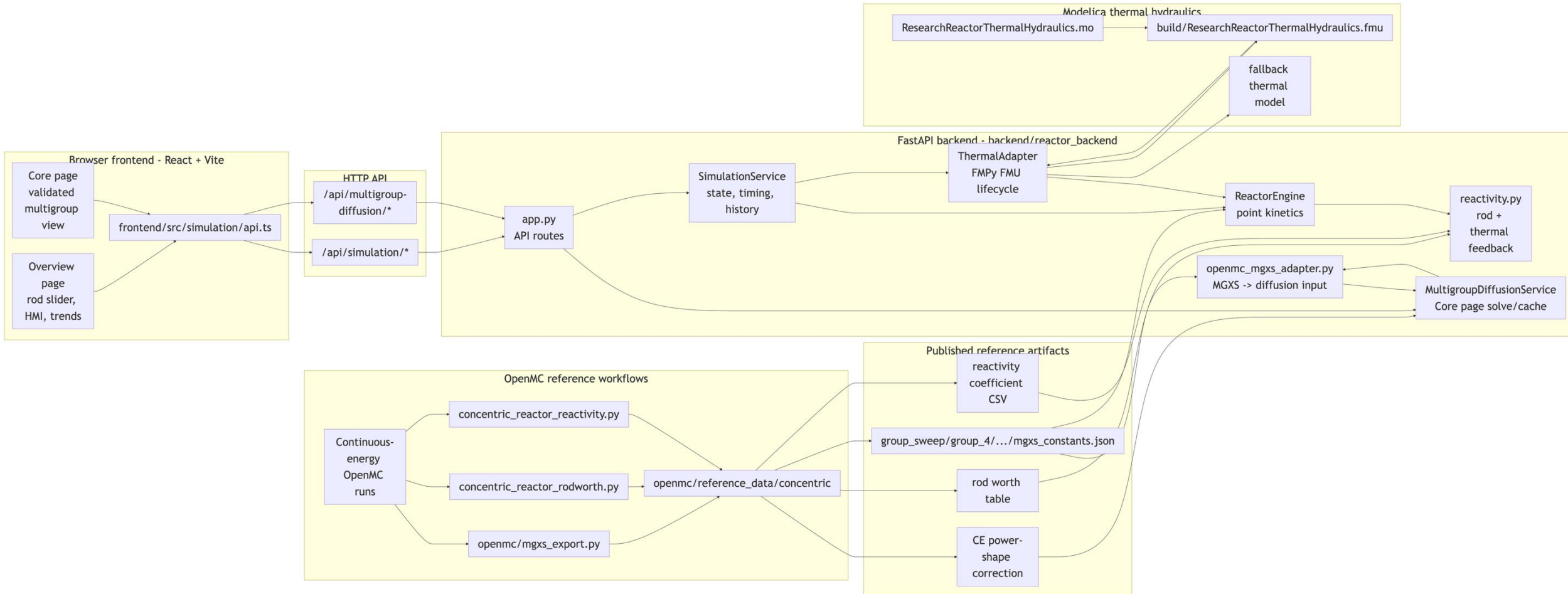


Pure post-solve CE power-shape correction



SANDBOX – WEB SIMULATOR

- Architecture (backend: modelica, python, fmpy | frontend: react)



Reactor Lab

Overview

Core

REACTIVITY

-0 pcm

TOTAL FLUX

 $1.07 \times 10^{12} \text{ n/cm}^2/\text{s}$

THERMAL POWER

14.3 MWth

SIM CLOCK

31m 52s

ROD



31.4%

SCRAM

Pause

Reset

PRIMARY COOLING LOOP — HMI VIEW



Neutronics



Thermal Hydraulics



Heavy-water moderated research reactor point-kinetics demonstrator with Modelica-coupled thermal hydraulics. The control rod bank adjusts reactivity; the primary D₂O cooling loop transports heat from the annular core through a heat exchanger. Six delayed neutron groups are solved implicitly in the Python backend.